

Bird: A Point Cursor for Virtual Immersive Environments

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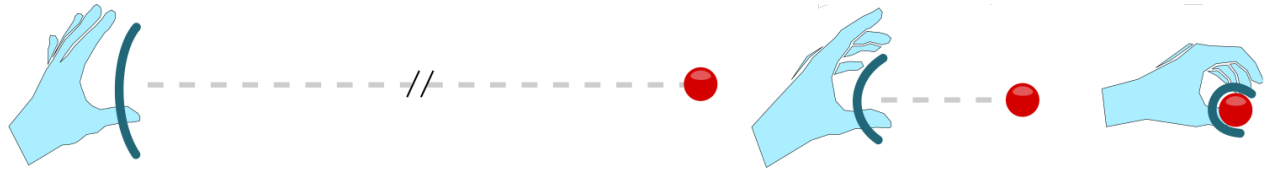


Figure 1: The Bird is a specific point in 3D space, which always remains some distance in front of the user's palm. Finger extension moves the Bird away from the hand, while closing the fingers brings it closer.

Abstract

This paper introduces the Bird, a novel point cursor for immersive virtual environments (IVEs) that enables precise, one-handed control over a point in 3D space beyond arm's reach. Interaction techniques commonly used in VR today lack this functionality. While direct manipulation allows for control of the position of an object in 3D space, it is limited to arm's reach. Ray-casting enables interaction at a distance but specifies a line rather than a point, making it impossible to move objects closer or farther without additional mechanics. The Bird overcomes these limitations by allowing users to select any visible object and place it anywhere within view, with one hand and without requiring a controller. We explore a range of use cases that highlight the Bird's potential to expand the design space for spatial computing.

CCS Concepts

• Human-centered computing → Pointing; Gestural input; Virtual reality.

Keywords

Virtual reality, 3D pointing, direct object manipulation, 3D user interfaces

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1 Introduction

Spatial computing is a bigger change than just a difference in hardware form factor. Immersive Virtual Environments (IVEs) are not particularly large screens. In IVEs, the user is inside of the interface, thereby giving users access to arbitrarily large spaces. While traditional “through the window” applications are still possible and familiar, the design space of spatial computing is much larger. Virtually limitless spaces require us to invent new design vocabulary that can make meaningful use of these boundless virtual landscapes.

A common means of interacting with objects in IVEs is direct manipulation. Direct manipulation is a technique that maps the user's real hand to a virtual representation of it in 3D space, and allows the user to interact with objects in a way that resembles interacting with physical objects in physical reality [9]. Because the virtual hand and physical hand are at the same location in direct manipulation, direct interaction techniques do not allow the user to interact with content beyond the natural working volume, or arm's reach without some variety of modification like that seen in Poupyrev et al. [10] or Bowman et al. [4].

Beyond the natural working volume, ray-casting techniques are common. The selection of objects using ray-casting involves casting a ray from a specific point in the virtual environment, such as the user's hand or a controller, into the scene. The user can then use this ray to point at objects. The first object hit by the ray is usually selected. Ray-casting techniques allow users to interact with objects at a distance quickly [8, 11].

However, because they describe a line, rather than a point, it is possible for ray-casts to intersect multiple objects, as shown in Figure 2. When this happens, ray-casts do not provide a way for a user to specify which object they want to select without an additional disambiguation step. An object selected with a ray-cast also cannot be moved along the length of the ray without some additional form of input, such as a button press. Ray-casts therefore do not afford moving objects closer to or farther away from the user. This means that, while a user stays standing in one place, ray-casts reduce a virtual environment, which contains an arbitrarily large 3-D space, into what is effectively a spherical screen.

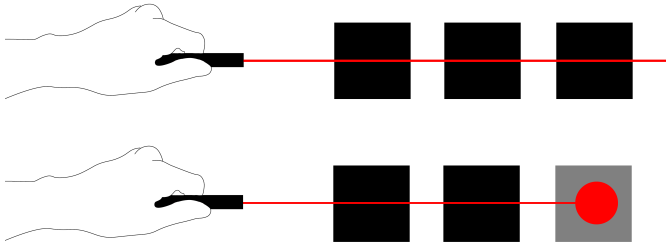


Figure 2: In the top of this diagram, a ray-cast is used to point to a series of three cubes. Because ray-casts describe a line, they provide no way for a user to specify which object along the ray they want to select without an additional disambiguation step if that line intersects multiple objects. The point cursor, shown in the bottom of the diagram, specifies an individual point along the ray. It is therefore possible to specify which of the objects along the ray should be selected.

Point cursor techniques resolve problems associated with ray-casts by specifying a point along a ray, as shown in Figure 2. The ability to move a point along a ray allows the user to specify which object they mean to select if the ray intersects multiple objects. It also allows users to select objects at different distances from themselves, and to move objects farther away or closer to themselves once they have been selected. This changes the nature of a user’s ability to interact with a virtual environment. Point cursors allow the user to specify and therefore interact with any point in 3D space within an arbitrarily defined radius.

This paper introduces the Bird, a novel point cursor technique. The Bird allows users to smoothly control and select a point in 3D space with one hand. The ability to smoothly control an arbitrary point in space with one hand opens new possibilities for how we interact with virtual immersive environments and spatial computing. This paper will present an exploration of the design space created by the Bird, along with four example use cases within this space.

2 Related Work

The two primary categories of egocentric methods for selecting and manipulating objects in virtual environments are virtual hand (direct manipulation) and virtual pointer (ray-casting) metaphors [11]. Ray-casting techniques allow users to interact with objects at a distance very quickly and precisely [8, 11]. However, because they describe a line, rather than a point, ray-casts provide no way for a user to specify which object they want to select without an additional disambiguation step if that line intersects multiple objects. Many techniques have tried to solve this problem by using separate disambiguation steps, heuristics, and additional inputs. Ray-casting techniques, because they specify a ray rather than a point, do not allow the user to bring an object closer to or farther away from them along the length of the ray. Direct object manipulation techniques can allow translation within the user’s natural working volume

using only one hand, but cannot reach beyond the natural working volume without some variety of abstraction.

The Bird is a point cursor (also referred to as a 3-D cursor [2]) which is a ray-casting technique that allows the user to specify a point along the ray. Point cursors allow the user to move selected objects closer to or farther away from themselves if the user is able to control the location of the point cursor along the ray. In one study, indirect depth control of the point cursor using buttons was shown to have a faster selection time than other methods [4]. However, the cursor moved at a linear rate along the length of the ray, which could become frustrating for users as they became more skilled with the technique. Other point cursors have introduced various mechanisms for controlling the position of the point along the ray. RayCursor is a technique that combines indirect point control of a point along a ray using buttons with a heuristic technique [3]. Jedi ForceExtension [7] moved objects along the length of the ray at a constant speed, using a combination of button press and head pose, but used only a standard ray-cast for selection. The Depth Ray is a ray cursor designed for use in volumetric displays [8]. The user can control the position of a point cursor along the length of the ray by moving the hand forward or backward, with a 1:1 control - display ratio. While this was a very effective technique in the small space of a volumetric display, it has not been tested in larger virtual immersive environments. iSith specified an individual point in 3D space by using two ray casts. One ray was controlled using each hand, and the point cursor was the point of intersection between the two rays [12]. iSith allowed users precise control in their natural working volume, and the ability to easily move objects which were farther away, but required two devices with 6-DOF tracking, and the use of both hands. Conductor is a similar technique which allowed users to specify a point in 3D space as the intersection between a ray held in the dominant hand and a plane held in the non-dominant hand [13]. While it was shown to be efficient to use, it still requires the use of both hands.

The Bird is uncommon among point cursors in that it allows the user to control the position of the point along the ray, in addition to the ray itself, with one hand, without the addition of other inputs such as gaze tracking or controllers, and can place the cursor at arbitrary locations, rather than snapping to predetermined points. Other techniques that meet these criteria are HOMER and Go-Go [5, 10]. The Hand-centered Object Manipulation Extending Ray-casting (HOMER) technique used ray-casting in order to place the cursor representing the user’s hand near the object they intended to select, and then directly manipulated that object at a normal scale, having effectively moved their local working volume to the location of the object [5]. When they let go of the object, the hand returns to the same location as its real-world counterpart. While this technique allows for direct object manipulation at a distance, because it incorporates a ray cursor, it still could not disambiguate between different objects intersected by the ray. It is also not a particularly useful tool for grabbing objects which are far from the user and efficiently pulling them closer, because a user’s movements remained on a one-to-one mapped scale, even when their hand was far away. Go-go is a technique that maps the hand non-linearly, such that a partially extended arm keeps the virtual hand and real hand in roughly the same location, but a more fully extended arm sends the hand much farther away [10]. This technique

allows users to interact with a larger sphere than their own arm span, while also retaining the control of direct object manipulation and without introducing any explicit mode switching. However, it leads to shoulder and arm fatigue due to its reliance on whole-arm motion. The Elastic Arm is a more recent adaptation of the Go-go technique [1]. When the arm is moved beyond a certain distance away from the body, the virtual hand moves away from the body in a rate-controlled way, rather than using a nonlinear mapping.

3 Bird Implementation

The Bird is a novel point cursor technique which allows the user to smoothly control and select a point in 3D space with one hand. The position of the cursor is along a ray drawn from the center of the user’s palm through the center of a sphere of best fit of tracked points of the user’s fingers, as shown in Figure 3. The distance between the center of the sphere and the center of the palm controls the position of the cursor along this ray. As the hand is opened, and the sphere made by the fingers is made larger, the cursor moves farther away. As the hand is closed, and the sphere is made smaller, the cursor moves closer to the hand, as shown in Figure 1. Subtle changes in the positions of individual fingers can be used to change the shape of the sphere of best fit, thus changing the direction of the ray. For example, curling the thumb towards the palm without moving other fingers will move the cursor both closer to the hand, and in the direction of the pinky. The index finger is not included in calculations of the sphere of best fit, and is reserved to click.

Implementation of the Bird requires hand tracking. It does not require eye tracking, controllers, or any other form of physical hardware. Use of the Bird requires the hand to be within the field of view of whatever system is being used for hand tracking. It does not require the user to hold their hand out in front of them. If the hand tracking system has sufficient field of view, the Bird can be used with the hand relaxed at the user’s side.

With regards to terminology, the Bird operates like a mouse. For a computer mouse, the term “mouse” refers to the physical hardware, the overall interaction technique, and the point specified on the screen. “Cursor” is used specifically to refer to the graphic representation of the point of interaction on the screen. In the context of the Bird, the term cursor is used to refer to the visible representation of the point of interaction in 3D space. Unlike the mouse, the Bird does not require a dedicated piece of peripheral hardware.

3.1 Sphere of Best Fit

We determine the sphere of best fit by using the linear algebra technique of least squares to minimize the squares of the distances between the tracked points of the hand and the sphere surface. We solve for c and r , where c is the center of the sphere and r is the sphere radius, such that the sum

$$\sum_{\text{points } \mathbf{x}} (d(\mathbf{x}, c) - r)^2$$

is minimized.

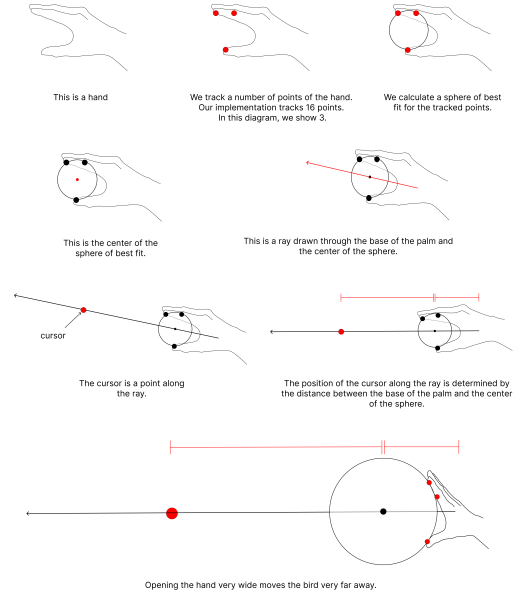


Figure 3: The position of the cursor is along a ray drawn from the base of the user’s palm through the center of a sphere of best fit of tracked points of the user’s fingers. The distance between the center of the sphere and the center of the palm controls the position of the cursor along this ray. As the hand is opened, and the sphere made by the fingers is made larger, the cursor moves farther away. The index finger is reserved to click.

3.2 Position of the Cursor along the Ray

The relationship between the radius of the sphere and the position of the cursor along the ray is nonlinear. While the cursor is near the user, it is approximately linear to enable the precise movement of small objects nearby. When the cursor is farther away, movements of the hand cause larger movements of the cursor, to allow for interaction with large objects at farther distances, such as moving architectural elements or trees. The final range R is calculated by blending between these two mappings based on the distance d between the palm and the sphere center c by defining two characteristic distances d_1 and d_2 , where $d_1 < d_2$, using the polynomial

$$P(d) = \hat{d}_1 + \hat{d}_1^2 + \hat{d}_2^5$$

where $\hat{d}_1 = d/d_1$ and $\hat{d}_2 = d/d_2$ are unitless normalized distances. The final range D is the distance d scaled by the polynomial blend factor, defined as

$$D = dP(d)$$

3.3 Jitter

Ray-casting techniques suffer from jitter [6]. Jitter is caused by the fact that a small point at the end of a long ray will magnify any unintentional motion or shakiness of a user's hand. We correct for this by increasing the degree of smoothing applied to the position of the cursor the farther it moves out from the hand.

We model the user's intended cursor position as a random process with variance Q corrupted by isotropic random noise with variance R , which increases with range. We estimate the intended position of the cursor using a Kalman filter. We scale R by the cube of the distance D^3 , which empirically gives good results.

3.4 Visibility at a Distance

As the cursor moves farther from the user, it occupies less of the user's field of view, and becomes more difficult to see. We compensate for this by increasing the scale of the cursor as it moves farther away. Increasing the size of the cursor also makes it easier to select small objects at a distance.

3.5 Click

To click, the user enters the tip of the index finger into the sphere of best fit. Entering the tip of the finger into the sphere is the equivalent of pushing the button of a traditional mouse down, and removing it again is the equivalent of releasing the button. To prevent rapid and unintended clicking when the tip of the user's index finger is close to the edge of the sphere, we define a distance that the tip of the pointer finger must penetrate the sphere to enter a selected state, and a distance out of the sphere that tip of pointer finger needs to be retracted to to exit selected state.

More details on our implementation can be found at its github repository: <https://github.com/SackJBones/bird-3d-cursor>

4 Design Space and Example Uses

The ability to smoothly control an arbitrary point in space with one hand opens new possibilities for how we interact with virtual immersive environments and spatial computing. There are four components of the Bird that may be most relevant to any particular interaction which builds on it. These **Interaction Components** (also shown in Figure 4) are:

- An **object** held by the cursor.
- The **cursor** itself, which is a point in 3D space.
- The **ray** extending from the hand to the cursor.
- The **path** drawn through space over time by the movement of the cursor.

We present examples of applications with mechanics that use each interaction component.

4.1 Held Object (Fruit Replacement)

To demonstrate an interaction in which the Interaction Component is the object held by the Bird, we created an experience involving an apple tree, shown in Figure 5. Users can pick apples off the tree using the Bird, and place them into baskets on the ground. This task could also be accomplished with ray-casting, as the placement location is predefined, and ambiguities could be resolved by selecting the nearest apple, since apples can be picked in any order. The ability

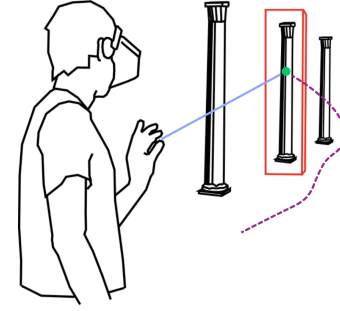


Figure 4: An interaction which builds on the Bird could be primarily centered around the object held by the bird (column highlighted in red), the cursor itself (shown in green), the ray extending from the hand to the cursor (shown in blue), or the path drawn by the cursor through space (the dotted purple line).

to control a precise point in 3D space is not required for this part of the interaction.

Once users have picked enough apples, they are presented with a basket of pumpkins and asked to place them on the tree. Pumpkins can be placed anywhere on the tree, and do not snap to predefined locations. This task requires the ability to control a specific point in 3D space, and could not be accomplished with a simple ray-cast.

4.2 Cursor (Fireball Demo)

To demonstrate an instance in which the Interaction Component is the cursor itself, we developed a simple game, in which the player uses the Bird to cast fireballs. Both hands control separate cursors, both of which have the same functionality, shown in Figure 6. The cursor can be passed through nearby chalices of fire, which causes it to “pick up” a fireball. When the cursor carrying a fireball collides with an object in the environment, the fireball explodes, and the cursor returns to its normal state. The environment contains several towers, and the goal of the game is to knock over the towers using the fireballs. Rather than throwing or launching the fireball, the Bird allows users to control the fireball's movement continuously until it explodes.

4.3 Ray (Pointing)

Pointing to a specific item in 3D space, as a means of showing it to another person, is an instance in which the ray drawn out from the hand is the Interaction Component of interest. By drawing the ray from the hand to the position of the cursor, the Bird can be used to simultaneously control the length and direction of the visible ray. To explore this concept and its practical implications, we present a hypothetical multi-user application which simulates bird watching. In this example, multiple users wearing HMDs share a virtual nature scene where 3D bird models fly between perches. Users are tasked with spotting birds and pointing them out to others using the ray emitted from their hand. This experience is inspired by the actual experience of bird watching, where people point with an extended arm to help others follow their line of sight. This task

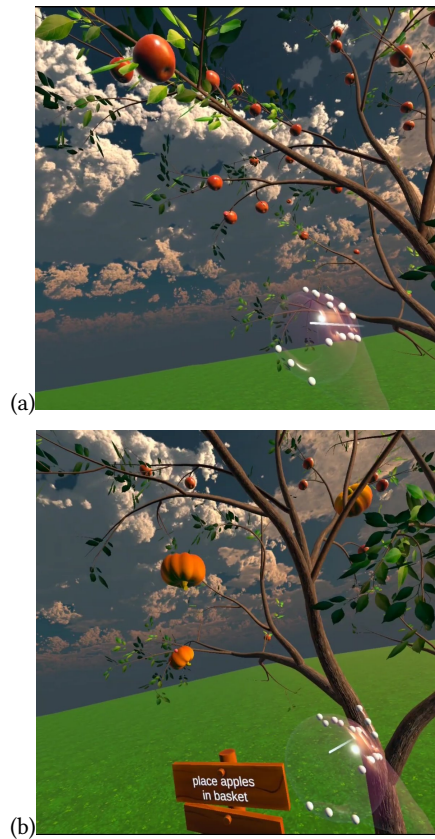


Figure 5: We demonstrate an application in which the object held by the Bird is the Interaction Component of interest with an experience involving an apple tree. (a) The Bird can be used to select any apple on the tree, including those which are far above the player, and place it in the basket below. (b) The Bird then can also be used to re-populate the tree with pumpkins, which can be placed at any location on the tree while still standing on the ground.

tends to involve real-world difficulty related to depth—determining if the bird is on a certain tree, or the tree behind it, can be difficult. The Bird resolves this difficulty by allowing the user to point to a specific position along the ray.

4.4 Path (Painting)

To demonstrate an instance in which the path of the cursor over time is the Interaction Component of interest, we propose a simple 3D painting application. In this example, the cursor would be used to draw a line following its path while the index finger is inside of the sphere of best fit, much like the drag interaction on a typical mouse. Properties of the line drawn by the cursor, such as its size and color, could be adjusted by passing the cursor through various menu options.

Large-scale 3D painting is an interaction that current techniques cannot fully achieve. Drawing paths larger than the user’s natural working volume is presently possible by sketching them at a smaller

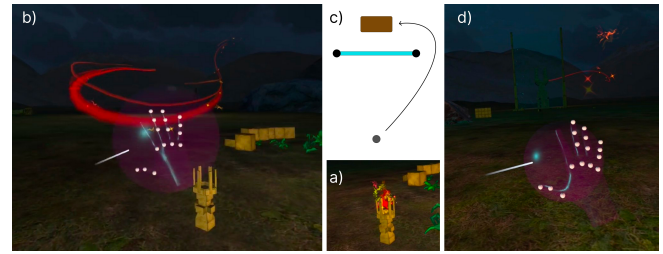


Figure 6: To demonstrate an instance in which the Interaction Component is the cursor itself, we developed a simple game, in which the player uses the Bird to cast fireballs and knock down towers. (a) The cursor can be used to pick up fireballs by passing it through various lanterns in the scene. (b) The Bird can be used to move the fireball to any visible location in 3D space, as can be seen in the glowing trail it leaves behind. (c/d) This control over a point in 3D space allows users to knock over a tower which is behind a shield, by moving the cursor around and behind the shield.

scale and then enlarging them, but this process is cumbersome and disrupts the fluidity of creative expression. Imagine an architect in the early design phase, standing inside a 3D model of a future building. With the Bird, they can seamlessly sketch expansive, gestural lines at full building scale, moving through the virtual space as they design. This approach enables natural, immersive interaction with large-scale virtual environments, a capability unavailable with traditional tools.

5 Discussion / Future Work

The Bird introduces new possibilities for interacting with virtual immersive environments by enabling precise, continuous control of a point in 3D space beyond arm’s reach. In this paper, we presented four illustrative applications: two implemented prototypes and two conceptual examples. These examples demonstrate a range of potential uses for the Bird, though its full range of possibilities has yet to be explored. Future work will focus on evaluating the Bird’s usability and effectiveness through user studies. These evaluations will guide the development of best practices for integrating the Bird into diverse virtual environments, and ensuring that it can be used effectively for real-world applications.

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